

METHODOLOGY

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20 years of PediSurv: a critical evaluation of a surveillance network for severe or rare infectious diseases in children in Belgium (2002–2023)

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Abstract

Background The Pediatric surveillance network, PediSurv, has been monitoring several rare or severe infectious diseases in children under 15 years in Belgium since 2002. The network collects clinical information in order to determine risk factors and evaluate vaccination strategies. Moreover, absence of disease is regularly reported to follow up international eradication and elimination targets set by the World Health Organization. Over more than 20 years surveillance, the importance and purpose of the network has evolved significantly in the growing landscape of surveillance systems, digital transitions and changing epidemiology. In this paper, we critically assess the strengths and weaknesses of PediSurv and explore its future role in the Belgian surveillance system.

Methods We conducted a descriptive analysis based on initial study protocols, quality documents and reports to reconstruct historical transitions of the network. Trends between 2002 and 2023 were analyzed for number of participants, coverage and contribution of registrations compared with other existing data sources, and complemented with findings from a 2022 user experience survey.

Results Digitalization and adoption of General Data Protection Regulation were transitions with the biggest impact on the network leading to stricter data governance requirements, including revised data collection procedures and increased administrative burden; this, in turn limiting data analysis and the flexibility of the network. Over 20 years, the number of participants gradually decreased. In parallel, case registrations decreased for most pathogens, especially relative to registrations in co-existing surveillance systems. However, in PediSurv, the degree of detail of clinical information remains high and it remains the unique data source to capture zero notifications. Participants cite importance for public health as the main motivation for participating.

Conclusion The surveillance landscape has drastically changed over the past 20 years, highlighting the need for existing systems to remain flexible and responsive to current and emerging public health needs. In the current Belgian context, whilst options for automatization are being investigated, manual reporting by physicians still offers

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added value. To ensure long-term engagement it is essential to sustain the motivation of participating physicians by clearly demonstrating the public health impact of surveillance systems and minimizing their reporting workload.

Keywords Disease surveillance, Severe disease monitoring, Pediatric infectious diseases, Vaccine-preventable diseases, Belgium

Text box 1: Contributions to the literature

- This manuscript describes how increasing digitization and data security regulations have impacted the structural organization and participation of a surveillance network.
- It highlights challenges in manual case reporting by physicians, such as limiting workload and maintaining motivation through frequent feedback.
- It emphasizes the importance of reevaluating and, if necessary, adapting existing surveillance systems in a rapidly evolving landscape.
- It provides insights that may be of interest to other national and international surveillance networks.

Background

Surveillance systems are essential components in the follow-up of infectious diseases. Since the introduction of vaccinations, the morbidity and mortality of different infectious diseases have been importantly reduced and worldwide at least 154 million lives have been saved over the past 50 years [1, 2]. Although the circulation of most diseases has decreased, high vaccination coverage and continuous surveillance remain necessary as outbreaks can still occur due to immunity gaps in the population [3]. In Belgium, infant vaccination coverage is high (> 90%) for all vaccines included in the free basic vaccination program [4]. The World Health Organization (WHO) has set several global targets for eradication of polio and elimination of rubella and measles [5, 6]. Follow-up of these targets is done in collaboration with stakeholders such as the Global Polio Eradication Initiative (GPEI) [7] and the European Centre for Disease Control (ECDC), who request reporting of key indicators from member states. In this context, the Belgian pediatric surveillance network, PediSurv, was established in October 2002 with the primary objective of following up polio, measles and rubella in children under 15 years in Belgium. In later years other diseases were added. Since then, parallel surveillance networks have emerged and continue to evolve. The network of sentinel laboratories was created in 1983 (SL) [8, 9] and hence already existed before the start of PediSurv. In contrast, mandatory notifications (MN) for selected diseases were only legally implemented from 2007 and are organised by the Flemish Region (Departement Zorg), the Brussels-Capital Region (Vivalis) and the Walloon Region (Agence Wallonne pour une vie de qualité-AVIQ) [10–13]. Finally, the network of national reference centres was founded in 2011 (NRC) [14, 15]. These parallel sources frequently complement each other by supplying new information,

but in some cases, they may overlap in information they contain. In this paper we describe the historical adaptations and evolutions of PediSurv between 2002 and 2023, within an evolving epidemiological and digital landscape, and assess whether the network's objectives have been met. This provides insights into both the current state of the network and potential opportunities, challenges and threats for the future that could be informative for similar systems elsewhere.

Methods

As a first objective we assessed the design of PediSurv and the reformations needed over time. Based on information retrieved from initial study protocols we described the design of the network and compiled the chronological changes in a detailed timeline. As a second objective, we looked at the evolution over time of the number of participants, the network's geographical coverage and the contribution of its registrations compared to other existing data sources. We placed these in perspective with key changes in the data collection methods. To calculate hospital coverage, we converted the number of participants to the number of participating pediatric hospitals. If at least one hospital-based pediatrician participated, the hospital was included. This enabled us to show the proportion of participating pediatric hospitals compared to the total number of pediatric hospitals in Belgium as registered by the Federal Public Service (FPS) Health, Food Chain Safety and Environment. For the geographical coverage we defined 'actively participating hospital' as a hospital with at least 1 pediatrician participating for a minimum of 6 months a year. By mapping the actively participating hospitals and combining this with their share of pediatric hospitalizations (< 15y) per municipality (as provided by the FPS [16]), we provide a coverage map showing, for each municipality, the percentage of hospitalizations covered by actively participating hospitals relative to the total number of hospitalizations. Lastly, we presented the findings of a 2022 survey conducted among participants, exploring their perceptions of the network and identifying potential areas for improvement.

Results

Description of the surveillance design and its transitions

Surveillance design

PediSurv monitors rare or severe infectious diseases in children under 15 years of age. Currently, the network

included surveillances of measles, mumps, acute flaccid paralysis (AFP), congenital rubella syndrome (CRS), invasive pneumococcal disease (IPD), and pertussis (case definitions of included pathogens are available in additional file 1). Participation is voluntary and participants are recruited among pediatricians (whole of Belgium) and general practitioners (exclusively for Brussels). The wider inclusion of participants for Brussels allowed closer monitoring of this particular area with a lower vaccination coverage at the start of surveillance. Participants are invited monthly to perform case registrations, or actively report the absence of diseases in the past month through a zero-case notification. Data is collected on socio-demographic variables, symptoms and complications, vaccination status, laboratory results, source of infection, risk factors, hospitalizations and treatments prescribed. The network is coordinated by Sciensano (Belgium's public health institute) and sponsored by the Flemish Region (Departement Zorg), the Brussels-Capital Region (Vivalis), the Walloon Region (Agence Wallonne pour une vie de qualité-AVIQ) and the FPS Health, Food Chain Safety

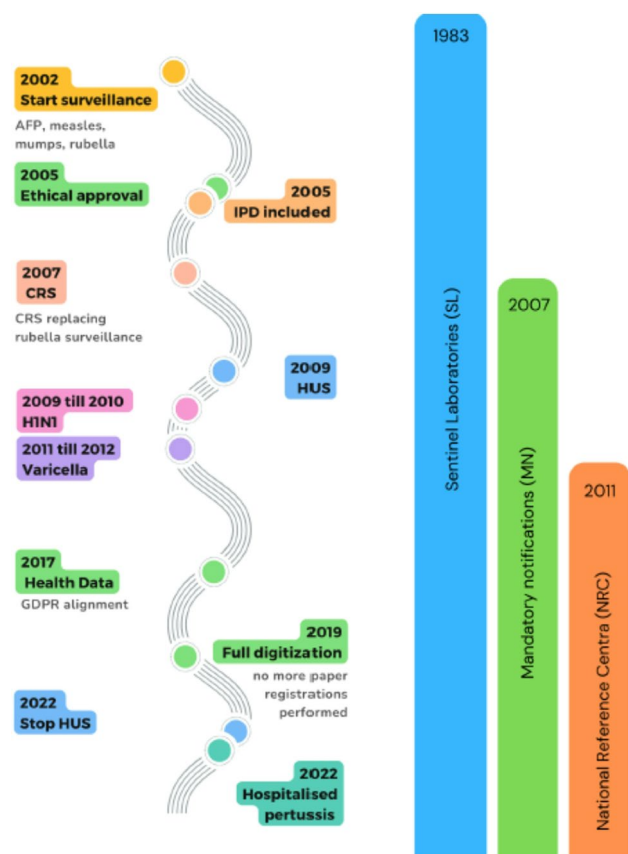


Fig. 1 Timeline of the Belgian surveillance network PediSurv, 2002–2023, with major changes to the network as well as existence of parallel surveillance networks. (AFP=Acute Flaccid Paralysis, IPD=Invasive Pneumococcal Disease, CRS=Congenital Rubella Syndrome, HUS=Hemolytic Uremic Syndrome, H1N1 = H1N1 influenza virus, GDPR=General Data Protection Regulation)

and Environment. Scientific guidance is received from a steering committee composed of academics, physicians from several key pediatric hospitals and policy makers.

Transition in included pathogens

The Timeline (Fig. 1) indicates the monitoring of Acute Flaccid Paralysis (AFP), measles and rubella since the start of the network in October 2002. Its primary purpose was international reporting and pursuing of WHO eradication and elimination goals. Mumps was included as it is part of the measles, mumps and rubella vaccine (MMR), and its lower vaccine efficacy makes it a useful early indicator to follow up the effectiveness of the MMR vaccine policy. In October 2005, Invasive Pneumococcal Disease (IPD) was added to monitor the impact of the conjugated pneumococcal vaccine, which was introduced in Belgium in 2004 and nationally implemented in 2007. Monitoring of Congenital Rubella Syndrome (CRS) replaced rubella surveillance in January 2007 due to the complex and inconclusive clinical presentation and serology of rubella cases. Hemolytic Uremic Syndrome (HUS) was added in March 2009 to better estimate its incidence in Belgium. Between 2009 and 2010 the pandemic influenza virus H1N1 was briefly monitored by a subset of pediatricians within the network to complement surveillance by the Sentinel General Practitioners (Sentinel GP) network [17] which did not sufficiently cover pediatric outpatients. Similarly, varicella was incorporated into the surveillance from 2011 until 2012 to make information available on the burden of varicella in Belgium and the possible implementation of universal varicella vaccination [18]. Data collection for HUS was discontinued in 2022 because of lack of added value compared to data already available through MN and NRC [19]. Finally, hospitalized cases of pertussis in patients under 3 years of age were added since the end of 2022 to monitor the severity and burden of the disease and to collect more detailed clinical and vaccination status information.

Transition in surveillance design

Requirements regarding data safety and required permissions changed drastically over time. At the inclusion of IPD in 2005, approval to collect detailed information was given by the ethical committee of Sciensano and the 'Queen Fabiola Children's University' Hospital in Brussels. In May 2016 the European General Data Protection Regulation (GDPR) was adopted and had to be legally integrated by member states by May 2018 [20]. As a result, PediSurv, requested and received approval of the Belgian information security committee in November 2016, under conditions designed to minimize data processing, limit duration of storage for maximum 30 years and ensure use exclusively for public health purposes. To be fully compliant with current and evolving

data protection legislation and standards, the network migrated to a new secure registration platform 'Healthdata' (HD) [21] (Fig. 1). Access to HD was limited to hospital-based physicians and initially restricted to a certain number of hospitals and only available through the hospital's Virtual Private Network. Since not all physicians worked in hospitals and the transition wasn't simultaneous across hospitals, the initial platform, managed by the department 'Epidemiology of Infectious Diseases' (EID) was also maintained, resulting in two parallel data collection flows. Physicians access either the EID application (for general practitioners and private pediatricians) or the HD application (for hospital-based pediatricians) through a personal login and password and register detailed case information. This data is pseudonymized and securely stored. Zero notifications are collected via monthly emails to all physicians, using a personalized encrypted link (Fig. 2). At the end of 2022, the technical hosting environment of the EID platform was changed, as a result of an institution-wide change in Sciensano. This resulted in a new log-in procedure for participants.

In 2024, the data security of the EID platform was reinforced and the login procedure was simplified by linking to a widely used secure digital identity application 'itsme'.

As of 2023, 46% of participants use the EID platform and 54% the HD platform. At the start of surveillance, all registrations were done on paper, sent by post, and manually entered into a secure dataset. Over the years the number of digital registrations gradually increased, initially thanks to a new simplified website in 2006 and reinforced by encouragement letters (with the principal actions in 2008 and 2010). Finally, in 2019, all participants were instructed to register exclusively through digital platforms (Fig. 3). The average monthly response between 2019 and 2023 was 66%.

Evolution of the number of participants, surveillance coverage and contribution of registrations

Number of participants

We display the evolution of the number of unique participants per year in addition to the number of participants who actively participate, i.e. at least 6 months per

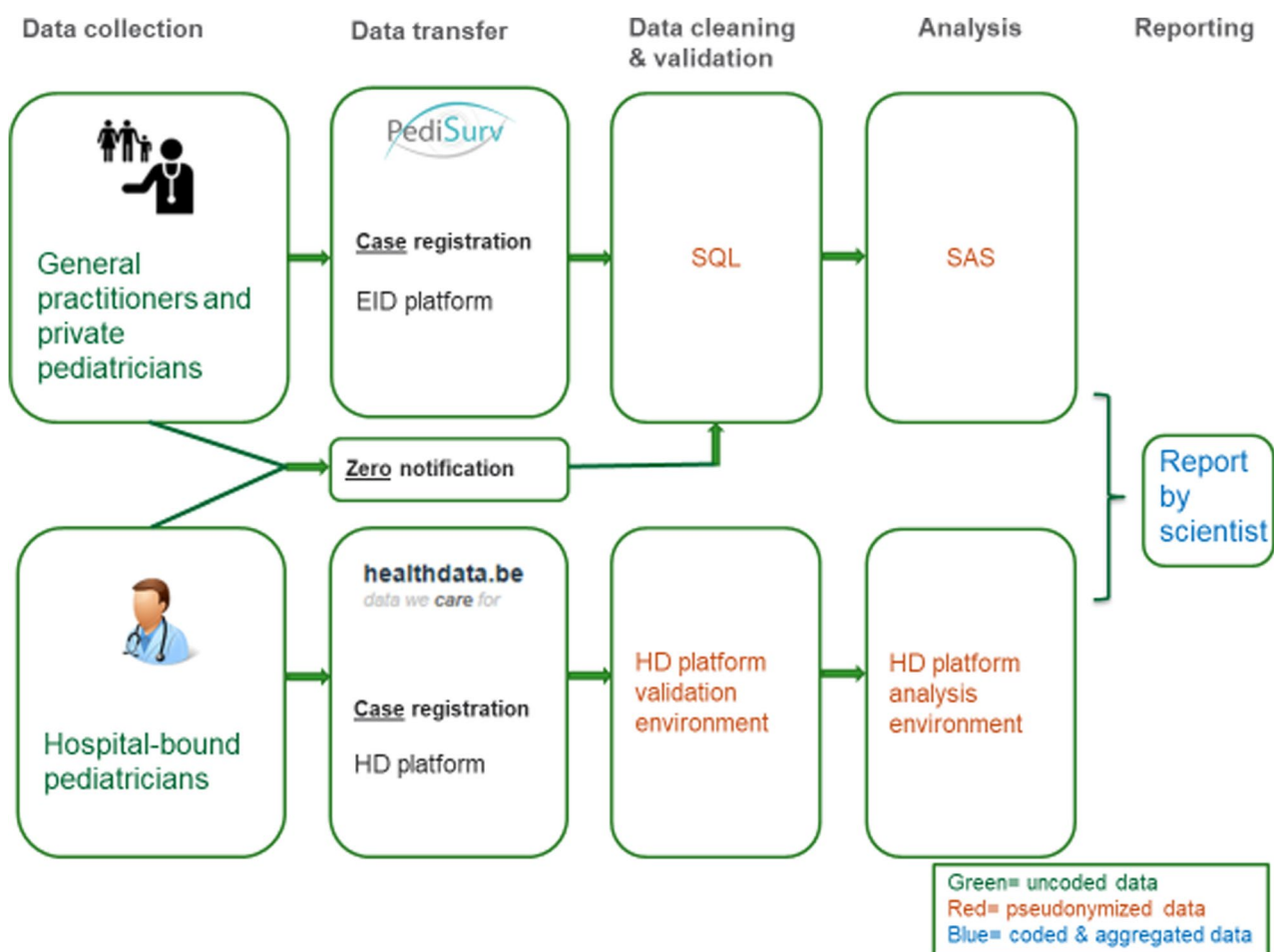


Fig. 2 Dataflow of the Belgian surveillance network PediSurv, 2002–2023. (EID= Epidemiology of Infectious diseases, HD=Healthdata, SQL= Structured Query Language, SAS= Statistical Analysis System)

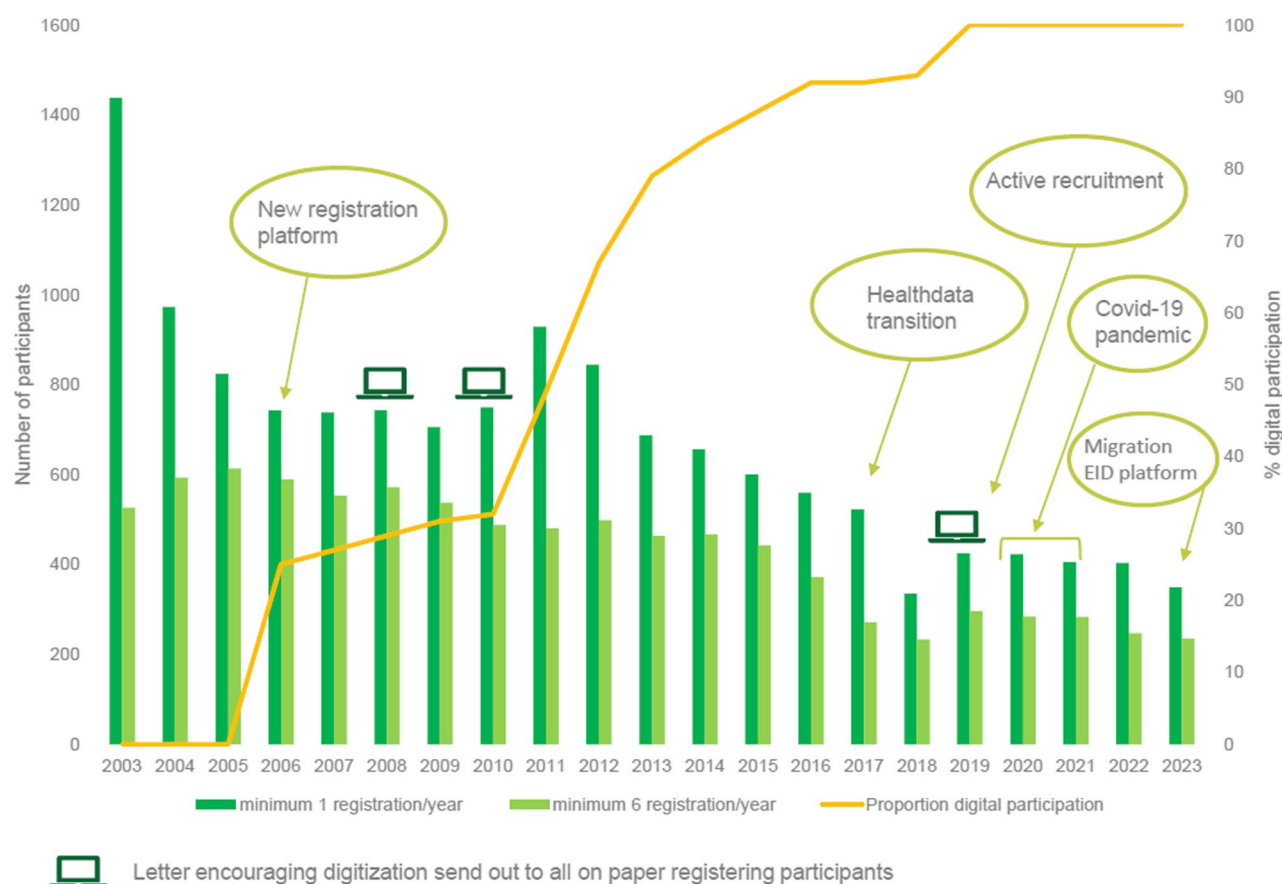


Fig. 3 Evolution of the number of physicians participating, and their digitization, in the Belgian surveillance network PediSurv, 2003–2023, with key influencing events indicated. (EID= Epidemiology of Infectious Diseases)

year (Fig. 3). Key events influencing participation are indicated above the graph. The number of unique participants declined from 1,437 in 2003, when all pediatricians in Belgium were actively invited to participate, to 349 in 2023. Looking at actively participating physicians, we see a more stable trend varying between 613 and 372 participants until 2016; followed by a sharp decline to on average 252 participants for 2017 and 2018. A limited increase to 296 participants followed in 2019 coinciding with active recruitment campaigns at targeted symposia, after which participation levels remained stable throughout the covid-19 pandemic years. The migration of the EID platform, necessary because of the institution-wide website migration at the end of 2022, resulted in a marked decline in the number of participating users in 2023. This was due to the requirement for participants to complete a complex and time-consuming re-registration process, introduced as part of strengthened cybersecurity measures within Sciensano. The process involved receiving and responding to multiple emails sent on different days (to confirm registration, provide personal details and the Belgian hospital accreditation number for hospital-based physicians, and set a new password). Not

all previously registered users completed this process, resulting in a decline in users. However, fortunately most of the users that were lost were already rarely participating to the network (non-active users). Looking at the proportion of digitally registering participants we see a clear jump after the second and third major incentive actions, but the switch to digital registration did not significantly impact the number of active participants.

Since the transition to HD, which is primarily accessible to hospital-based pediatricians, the focus has shifted more toward hospitals rather than individual participants. This shift is also driven by the preference of some participating hospital-based physicians to centralize registrations within the same hospital, as well as by the evolution of the included pathogens, which mainly require hospital care. Consequently, achieving a high coverage of hospitals is more important than maximizing the number of individual participants. Therefore, we also show the yearly total number of hospitals with pediatric beds in Belgium (provided by FPS) and their proportion participating in PediSurv (Fig. 4). With exception of the period 2017–2018, coinciding with the launch of the HD platform, the number of participating hospitals and potential



Fig. 4 Evolution of the number of pediatric hospitals and the proportion participating to the Belgian surveillance network PediSurv, 2003–2023. (Source: Federal Public Service, PediSurv)

coverage of pediatric hospitalizations is remarkably stable. In 2003, 69 hospitals participated, covering 66% of pediatric hospitals, compared to 67 hospitals in 2023, covering 74% of pediatric hospitals. The decreasing total number of pediatric hospitals is due to structural changes such as mergers between hospital.

Geographical coverage of the network

The map provides an overview of the theoretical coverage of the network (Fig. 5). Overall, coverage is good in Flanders and Brussels, both located in the northern part of Belgium (respectively 74% and 71% in 2023), but sub-optimal in Wallonia, the southern part of Belgium (48% in 2023).

Contribution of registrations

Case reporting (including zero notifications) in PediSurv have declined significantly and other surveillance systems currently play the leading role in monitoring, due to more complete case ascertainment. Nevertheless, the network continues to make a meaningful contribution (Table 1 presents a comparison of the purpose, aims and data provision of PediSurv and other relevant surveillance systems). For example, while the MN and NRC currently provide exhaustive surveillance for measles; and SL and (likely soon) the Sentinel GP network

offer the most complete data on mumps, PediSurv was initially the primary data source for both diseases [22, 23]. Thanks to its longer existence and established experience of its researchers, PediSurv appears to have contributed to developing expertise within the team, which has subsequently supported the setup of future surveillance systems. For IPD, the NRC is currently the most comprehensive source of data. However, PediSurv still offers an important supplementary safety net by providing detailed information on vaccination status and clinical characteristics, which enhances the overall dataset. And while pertussis cases in children under 3 years of age are captured through the MN, PediSurv adds unique and valuable insights concerning symptoms, comorbidities, hospitalization (including duration), intensive care unit admissions, treatment, outcome, and vaccination status (mother and child); data not captured by other systems, but crucial for in-depth analysis. Notably, for AFP, PediSurv remains the primary source of apportioning case reports since the inception of AFP surveillance.

Moreover, PediSurv remains to date the only national source in Belgium systematically providing zero notifications for children under 15 years of age. These data are essential for maintaining the country's rubella [24] and polio [25] elimination status by supporting the absence of cases, and also serve, in the context of measles

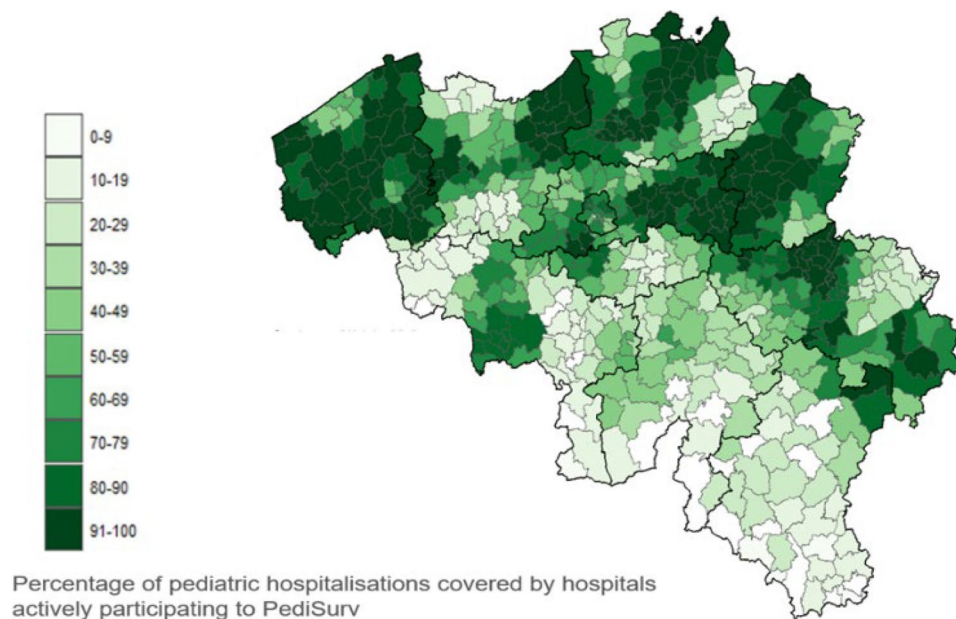


Fig. 5 Percentage of pediatric hospitalizations per municipality covered by hospitals actively participating in the Belgian surveillance network PediSurv, 2023. (Source: Federal Public Service, PediSurv)

Table 1 Comparison of purpose, aim and data provision by different data sources for epidemiologic surveillance in Belgium, 2023

	PediSurv	Mandatory Notifications	National Reference Centres	Sentinel Laboratories
Purpose	Contribute to elimination goals and in-depth analysis on selected pathogens.	Prevention of infectious disease spread in the community.	Microbiological monitoring of specific pathogens.	Rapid trend monitoring of disease incidence.
Aim	To collect zero notifications and detailed clinical information, often not captured by other data sources.	To collect contact and clinical information from infected patients and their close contacts.	To collect detailed information on pathogen evolution (genotyping, antibiotic resistance,...).	To collect number of positive cases from subset of laboratories.
Data provision	Voluntary participation of clinicians, who manually report.	Mandatory reporting by clinicians and lab technicians of confirmed (or suspected) case to the regional health authorities.	Designated reference laboratories, funded and responsible for monitoring specific pathogens, provide automated data extraction from laboratory database.	Voluntary participation of laboratories, with automated data extraction from laboratory database.

surveillance, as a complementary data source supporting the absence of measles when case numbers are low. Although zero notifications are restricted to children under 15 years of age, cases would be expected to occur primarily in this population in the event of transmission or reintroduction (for example, congenital rubella syndrome is mostly diagnosed within the first year of life), thereby providing evidence for the absence of endemic circulation. Number of zero notifications decreased over years, in line with declining participation as mentioned above, but also influenced by number of case registrations (e.g. high numbers of measles and IPD notifications in 2011 caused the overall zero notification rate to drop) (Fig. 6). From 2025 onward, transitioning to zero reporting per pathogen should improve the accuracy and utility of the data.

Survey results

Demographic data on participants is not routinely available. An ad-hoc survey conducted in 2022 and completed by 131 of participants spread over the different Belgian regions (35% response rate) showed that the majority of participating physicians are in older age groups (66% between 50 and 69 years and 10% over 69years, meaning past retirement age) [26]. All participants surveyed considered PediSurv important for public health and 82% indicated that a financial payment would not be an additional motivator to participate. However, improved feedback and a more user-friendly registration system would boost participation motivation for respective 62% and 39%. In response, since 2023, improvement of active

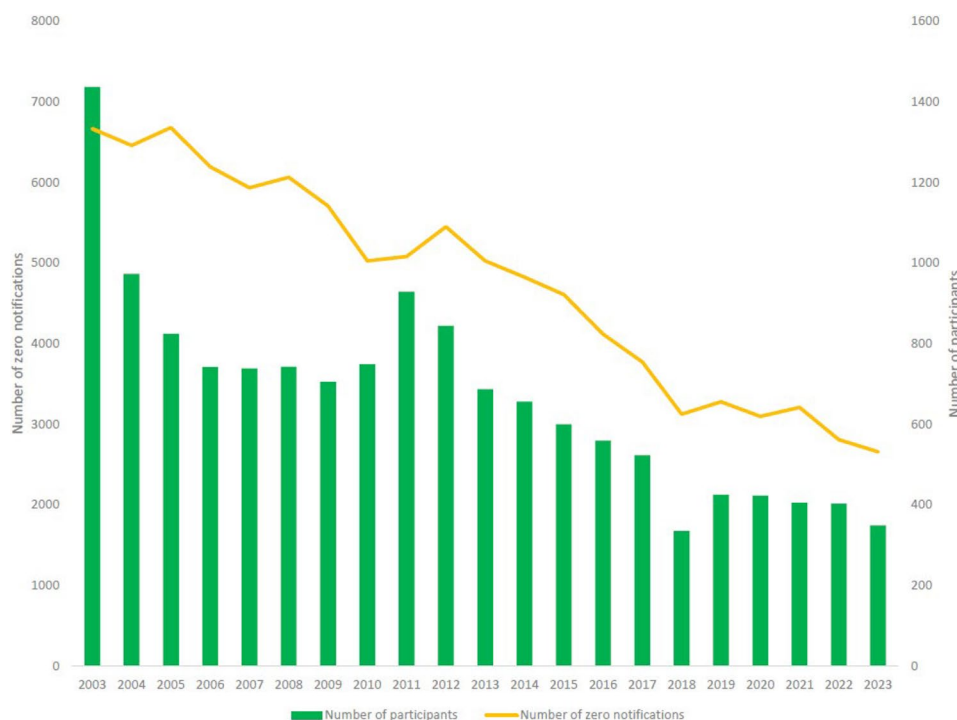


Fig. 6 Evolution of number of zero notifications in the Belgian surveillance network PediSurv, 2003–2023

communication and diffusion of results was performed by a quarterly newsletter, an annual report and continued presentations or posters at targeted symposia with high pediatrician attendance. An updated EID platform is available since January 2024 with a simplified login and usage.

Discussion

In this paper, we critically assess the strengths and weaknesses of PediSurv and explore its future role in the Belgian surveillance system.

Between 2002 and 2023 the surveillance landscape evolved, several new data sources were established and overall quality of data collection improved. The level of detail and completeness of data varies by source and purpose. MN focuses on rapid action to prevent further infection spread, prioritizing speed and identifying at-risk contacts, with information collected by regional health inspectors through physicians and patients. NRC's main goal is to microbiologically monitor pathogens for mutations and resistance and to evaluate whether currently used vaccines and antibiotics remain appropriate, while SL quickly tracks epidemiological trends without in-depth data collection. NRC and SL data are extracted from laboratory databases and provided to Sciensano. PediSurv does not have the ambition to collect all cases or quickly detect trends, but does collect detailed information often missing in other sources, essential to initiate analyses on the effectiveness of vaccination (i.e.

IPD, Varicella, H1N1) and to determine patient profiles and risk factors. Initially functioning as a major (and often the only) data source for incidence estimates and clinical information on various pathogens, the network has evolved into an additional safety net for obtaining detailed info on clinical information and vaccination status. In this way, PediSurv currently complements already available information with the goal of having as complete data as possible for (inter)national, regional reporting and further research (for example, by completing missing vaccination status of IPD cases for vaccine effectiveness studies [27]). In addition, it remains the unique source for zero notifications, key to prove sufficient surveillance quality and maintain elimination or eradication status. Challenges for the future are to continuously evaluate the (possible) added value of surveillances in the larger picture (Table 2 provides an overview of the principal challenges and opportunities for PediSurv).

In addition to the increasing number of surveillance systems, the growing digitization and the implementation of GDPR have also had a major impact on the functioning of the network, resulting in structural changes to data flows and higher data security standards. These severely limit the flexibility and simplicity of the surveillance and increases complexity for participating physicians (e.g. by changing log-in procedures) and involved scientists. Rapid inclusion of new diseases or additional questions in response to an emerging threat or availability of a new preventive measure has been an advantage of PediSurv

Table 2 Opportunities and challenges for PediSurv, the pediatric surveillance network for severe or rare diseases

Opportunities	Challenges
Increase maximal collaboration between data sources through automated data extraction and linkage of administrative datasets.	Continuously evaluate the added value of PediSurv within the overall surveillance landscape.
Valuable network of motivated clinicians willing to contribute to public health.	Keep (young) clinicians aware of the added value of PediSurv through regular communication of results in the form of scientific output.
-	Enhance PediSurv's flexibility while ensuring compliance with current data protection rules.

in the past. For example, for the temporarily follow-up of H1N1 from 2009 to 2010, PediSurv was a dynamic and flexible tool that quickly recruited a large number of physicians willing to register a new emerging infectious disease [28]. PediSurv’s flexibility was truncated by mandatory, time-costing set-up procedures such as approval by the information security committee, implementation of the technical changes in the HD data collection platform (with lots of competing priorities), and the necessity to carry out a small cell risk analysis to exclude risk for identification of the patient by the researcher. This led to missed surveillance opportunities, e.g. in the occurrence of hepatitis of unknown origin in children post-covid and RSV surveillance in the light of the impending immunization and vaccination start. An important opportunity for enhancing collaboration between data sources lies in integrating automatic extraction of data from electronic health records and linking administrative datasets. However, several legal and logistical obstacles still need to be overcome to enable more efficient, comprehensive and timely surveillance in the future.

The number of participating physicians experienced a large decline over the years, and with it the number of registrations. We attribute this to several factors, the most important factors, the rise in surveillance activities and the advent of the GDPR, are discussed above. Although the number of participants declined strongly, it is important to note that coverage of pediatric hospitals remained stable. A regional imbalance with underrepresentation in Wallonia is seen, although most major pediatric hospitals are involved. PediSurv constitutes a powerful network of motivated, voluntary participating physicians who would be willing to respond in case of newly emerging infectious diseases (if current legislations and applications allow).

Within the Australian Pediatric Surveillance Unit (APSU), part of the ‘International Network of Paediatric Surveillance Units’ (INoPSU [29, 30]), which currently includes approximately ten different countries (including PediSurv) with active pediatric surveillance units, similar

issues to those described above have been reported (An overview of the affiliated networks is provided in Additional file 2). A comprehensive review article on the APSU reported that the unit maintained approximately 1500 participating physicians between 1993–2019, with a monthly response rate of > 90% [31]. However, since the covid-19 pandemic participant numbers has declined and the response rate has dropped to 80% and lower, due to clinician fatigue and ongoing disruptions [32]. In PediSurv very little impact on number of participants was seen during the covid-19 pandemic, but the general average monthly response rate was much lower, at 66%, between 2019–2023. This difference may be attributed to the more limited overlap of surveillance sources in Australia compared to Belgium. More selective inclusion of pathogens in PediSurv could possibly benefit its response rate.

A survey conducted among APSU contributors in 2007 revealed findings similar to those of the PediSurv survey. Namely, that the majority of respondents viewed the network as valuable and recognized its potential to contribute to research [33]. The same areas for improvement were identified, in particular the need for more active communication of results, with the challenge on reaching a dynamically evolving group of pediatricians. However, improved frequency and quality of communication on PediSurv results since 2023 have not yet translated into a rise in participations. Active involvement of the national pediatric training programs and the professional associations to continuously inform and invite graduating paediatricians on the network, as in Australia, could possibly help address this. Interestingly, the majority of PediSurv survey respondents indicated that financial compensation would not provide additional motivation for their participation.

Given the extensive time span covered in this study, the varying levels of researcher involvement throughout the network’s development, and the inconsistent historical documentation, reconstructing a clear and accurate overview of the network’s evolution and the rationale behind key decisions proved to be a challenge. Additionally, the data (originally gathered from diverse sources) was manually merged in earlier stages, further complicating efforts to trace its exact origins. Lastly, the coverage calculations include only hospital-based physicians and assumes that pediatricians in actively participating hospitals report all cases. This approach generally overestimates coverage. Despite these challenges, the analysis of the network’s evolution has significantly deepened our understanding of its historical and current value, as well as the motivations behind critical decisions and transitions over time. As many international networks seem to be facing similar issues, our detailed documentation and analysis over an extended time frame can help to make PediSurv as well as other networks future-proof.

Conclusions

The surveillance landscape has drastically evolved over the past 20 years, underscoring the need for existing systems to remain flexible and responsive to current and emerging public health needs. In Belgium, despite major evolutions in the role and format of PediSurv, this analysis helps to reveal that the network of manual reporting by physicians remains an essential data source. Future opportunities to enhance surveillance include increasing automatized data collections and data linkage projects. Moving forward, the key challenge will be to ensure that diverse surveillance systems are well-integrated to maximize their public health value, while minimizing the burden on participating physicians so that they remain motivated for further participation.

Abbreviations

WHO	World Health Organization
GPEI	Global Polio Eradication Initiative
ECDC	European Centre for Disease Control
MN	Mandatory Notifications
NRC	National Reference centre
FPS	Federal Public Service
CRS	Congenital Rubella syndrome
AFP	Acute Flaccid Paralysis
MMR	Measles, mumps and rubella
IPD	Invasive Pneumococcal Disease
HUS	Hemolytic Uremic syndrome
Sentinel GP	Sentinel General Practitioners
HD	Health Data
EID	Epidemiology of Infectious Diseases
SL	Sentinel Laboratories
GDPR	General Data Protection Regulations
ICU	Intensive Care Unit
INoPSU	The International Network of Paediatric Surveillance Units
NSCK	Nederlands Signaleringscentrum Kindergeneeskunde
APSU	Australian Pediatric Surveillance Unit

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13690-026-01864-9>.

Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

CD, IP, MF and YD performed data curation. CD, IP, and MF conducted the analyses. IP was the major contributor in writing the manuscript. All authors critically reviewed, contributed to, and approved the final version of the manuscript.

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Data availability

The data obtained through the surveillance are available from the corresponding author on reasonable request according to Sciensano scientific policy.

Declarations

Ethics approval and consent to participate

Collection of data is allowed by the ethical committee of Sciensano and the 'Queen Fabiola Children's University Hospital, Brussels' and the commission for the protection of the privacy (last consultation can be found here). A letter of information for the patient, the storage period, the right to examine and correct own data and the contact details of the person to contact to exercise those rights was provided to all physicians to make it visible to patients. All data retrieved by the surveillance are saved on the secured server of Sciensano or the secured Health Data platform, depending on the used dataflow.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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